

Section D: Define study time periods					
Study Time Period	Label	Definition	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
10 STUDYPD		01 July 2015 - 30 June 2025 (inclusive)	Full study time period		OK19012026
11			Allows for 6mos lookback and no required minimum length of follow-up. ID_PD end at ends at the earliest of death or the end of the study period.		
12 ID_PD	Patient identification period	01 Jan 2016 - 30 June 2025 (inclusive)	Patients are required to have at least 1 inpatient cancer visit for multiple myeloma (ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) or at least 2 outpatient visits in any position on claim, on separate days within 90 days, during identification period. (Patients with 2 claims 30 or 60 days apart will also be flagged for consideration.)		OK19012026
13 INDEX	Index date	Date of the patient's earliest qualifying multiple myeloma diagnosis date	The index date is the date of the first medical claim for multiple myeloma. Diagnosis codes for MM are in tab 42. "MM dx code". Flags should also be made to distinguish which claims are non-diagnostic claims. A non-diagnostic claim refers to a claim for treatment of a condition (e.g., infusion) rather than diagnosis (see tab 49. "Diagnostic"). Patients must have six months of CE with medical or pharmacy benefits before the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled		OK19012026
14 Baseline	Baseline period	The baseline period is fixed and is the six months before the index date.	Patients are only required to have ≥ 1 day of CE starting on the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled; consider medical and/or pharmacy benefits. Follow up period ends the earliest of death or the end of the study period (30 June 2025)		OK19012026

A	B	C	D	E	F	G
FU_PD	Follow up period	The follow up period is fixed and is from the index date til the end of the patient identification period (inclusive).	Patients are only required to have ≥ 1 day of CE starting on the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled, consider medical and/or pharmacy benefits. Follow up period ends the earliest of death or the end of the study period (30.June2025). Note: Patients with ≥ 3 months of CE with medical or pharmacy benefits from the index date (with no allowable gaps in enrollment) will be flagged for consideration for sensitivity or secondary analyses, and ending on the earliest of health plan disenrollment or the study period for patients who do not have have an assigned date of death < 3mos after the index date.		OK19012026	
e.g. PST24	e.g. 24-month post-index period	e.g. INDEX_DT through INDEX_DT +730				
etc						

Section C: Source Data Preparation

Instruction: Section C should ONLY be used to document decisions to the source data before the analytic file build. If decisions apply to a variable in the analytic file, document in the appropriate variable definition.

#	Source Table/Variable	Situation Requiring Resolution	Action	Reasoning / Comments	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
1	Days to months conversion	Assuming 30.4375 day per month	3 months = 91 days, 6 months=183 days, 9 months=274 days, 12 months=365 days	Consistent with derivation other GSK Oncology studies		
2	ELUG_COH	Study team does not want to exclude smoldering MM patients. Optum built study cohort with the intention to exclude smoldering MM patients.	The team will additionally include patients who had ≥ 1 non-diagnostic medical claim for MM during the baseline period. These patients were excluded by Optum originally. If a patient has newly diagnosed or active MM, it would be likely that they would experience frequent activity, hence why we look for 2 non-diagnostic MM claims close together in the first step of the attrition table. A smoldering patient would have infrequent activity, hence why a claim for MM that falls outside of the 30, 60 or 90 days window requirement that's used to identify newly diagnosed patients is likely a smoldering patient.	Study team will consider excluding patients with a stand alone treatment code (rather than any non-diagnostic code) on an MM claim in the baseline period for future sensitivity analysis. This would not remove smoldering patients but rather ensure our patients were newly active MM patients.		OK19012026
3	FU_PD	Original IE criteria from Optum only considers patients with at least 6 mos of CE from index onwards.	Study team wants to keep criteria as broad as possible.	Will only require at least a day of CE post index, though will flag patients with 3 or more months of CE, to potentially subset on these patients for later consideration.		OK19012026

Approach: We implement both Approach 1 and Approach 2 in combination (OR logic) per Optum CDM business rules guidance:

Approach 1 (POSTOS): Inpatient if POS IN (21, 51, 61) OR TOS_CD IN ('FAC_IP.ACUTE', 'FAC_IP.REHSNF', 'PROF.INP.VIS', 'FAC_IP.SNF')

Approach 2 (CONF_ID): Inpatient if CONF_ID is NOT NULL and validated against T_CONFINEMENT table with both

ADMIT_DATE and DISCH_DATE present

Non-inpatient: All records not meeting either approach above

A claim is classified as inpatient if either approach identifies it as inpatient. This maximizes capture of true inpatient encounters.

Rationale: Using both approaches together provides the most comprehensive inpatient identification. From our validation on the 2025q2 data:

Total patients (Step 0):	94951
Any inpatient event:	43439 (45.2%)
Via POSTOS (Appr 1):	42618 (44.3%)
Via CONF_ID (Appr 2):	40357 (42.0%)
Both approaches:	39536 (41.1%)
POSTOS only:	3082 (3.2%)
CONF_ID only:	821 (0.9%)
Outpatient only:	52757 (54.8%)

